

Swimming Facility Inspection Outcomes in Toronto*

A Descriptive Analysis of Facility Types and Recorded Infractions

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By examining how inspection outcomes and recorded infraction patterns vary across different types of swimming facilities in Toronto using publicly available SwimSafe inspection data, this analysis aggregates 17,891 raw records into 9,967 inspection-level observations and compares Indoor Pools, Outdoor Pools, Spa Indoor facilities, and Spa Outdoor facilities. The results suggest that Outdoor facility types have higher observed proportions of non-Pass inspection outcomes than Indoor Pools, while the composition of recorded infractions also differs across facility types.

1 Introduction

Public swimming facilities provide important recreational spaces, but their operation also involves potential risks related to water quality, chemical exposure, and other environmental and operational conditions. The World Health Organization identifies swimming pools and similar recreational-water environments as settings in which these hazards need to be controlled and monitored (World Health Organization 2006). The Centers for Disease Control and Prevention similarly provides guidance for public aquatic facilities covering areas such as water testing, facility operation, maintenance, and inspection (Centers for Disease Control and Prevention 2024a).

In Toronto, the SwimSafe program is used to inspect public swimming facilities and make inspection information available through Toronto Open Data (Toronto Public Health 2026). The dataset contains information about inspection outcomes as well as recorded infractions, including their categories and types. These records provide an opportunity to examine whether inspection outcomes and the patterns of recorded infractions differ across types of swimming

*Code and data are available at: https://github.com/rallan-vijval/Toronto_SwimSafe

facilities, building on the broader role of inspections in monitoring public aquatic facilities (Centers for Disease Control and Prevention 2024b).

This paper asks: **How do inspection outcomes and recorded infraction patterns vary across different types of swimming facilities in Toronto?** To answer this question, I aggregate the raw SwimSafe records to the inspection level and compare Indoor Pools, Outdoor Pools, Spa Indoor facilities, and Spa Outdoor facilities. I examine inspection outcomes, the distribution of recorded infraction types, and the categories and specific deficiencies associated with those infractions.

The analysis shows that inspection outcomes and recorded infraction patterns vary across facility types. Outdoor facility types have higher observed proportions of non-Pass inspection outcomes than Indoor Pools, while the composition of recorded infractions differs across facility types. Comparing facility types is useful because the types and frequencies of recorded inspection findings may differ across facilities with different operational characteristics, making aggregate inspection counts alone difficult to interpret.

The remainder of the paper first describes the SwimSafe data and the construction of the inspection-level analysis dataset. It then presents the geographic and temporal coverage of the records before examining differences in inspection outcomes and recorded infraction patterns across facility types. The paper concludes by discussing the main findings and limitations of the analysis.

2 Data

The analysis uses the City of Toronto’s SwimSafe dataset, which contains records from inspections of swimming facilities in Toronto (Toronto Public Health 2026). The dataset used for this analysis contains 17,891 records covering inspections dated from July 8, 2024 to September 25, 2026. The raw data include information about the facility and establishment, inspection date and status, and recorded observations and infractions.

2.1 SwimSafe Inspection Data

The raw dataset contains multiple records for some inspections because a single inspection can include multiple observations or recorded infractions. Therefore, a row in the raw dataset does not necessarily represent one complete inspection. For this analysis, an inspection is defined as one establishment, one facility type, and one inspection date. Records were grouped using these three variables and aggregated to create an inspection-level dataset. This produced 9,967 inspections from the 17,891 raw records. The analysis dataset contains 9,967 inspections across four facility types. Indoor pools account for the largest number of inspections, while outdoor spas account for the fewest. The difference in the number of observations across facility types

Table 1: Example observations from the SwimSafe inspection data

Facility	Facility type	Inspection status	Inspection date	Infraction type
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-06-04	None
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-03-11	None
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-01-27	M - Minor
YORK RECREATION CENTRE	Indoor Pool	Pass	2026-01-27	M - Minor
YORK RECREATION CENTRE	Indoor Pool	Pass	2025-11-07	None

Table 2: Number of inspection-level observations by facility type

Facility type	Number of inspections
Indoor Pool	5,265
Outdoor Pool	1,463
Spa Indoor	3,041
Spa Outdoor	198

is important when interpreting comparisons, particularly for outdoor spas. Table 2 shows the number of inspection-level observations available for each facility type.

Because the number of inspections differs across facility types, comparisons are generally presented as percentages within each facility type rather than as raw counts alone. This allows the distribution of outcomes and recorded infractions to be compared across groups of different sizes. The relatively small number of Spa Outdoor inspections is considered when interpreting the results. The analysis was conducted in R (R Core Team 2026) using the `tidyverse` collection of packages (Hadley Wickham and Mara Averick and Jennifer Bryan and Winston Chang and Lucy D’Agostino McGowan and Romain François and Garrett Grolemond and Alex Hayes and Lionel Henry and Jim Hester and Max Kuhn and Thomas Lin Pedersen and Evan Miller and Stephan Milton Bache and Kirill Müller and Daniel S. Woo and Hiroaki Yutani 2019), `ggplot2` (Wickham 2016) for visualization, `tinytable` (Vincent Arel-Bundock 2026) for tables, the `opendatatoronto` package (City of Toronto 2026) to access and process the Toronto Open Data dataset, and `here` (Kirill Müller 2026) for project-relative file paths.

2.2 Geographic and Temporal Coverage

The SwimSafe records also provide geographic and temporal context for the inspection data. The dataset contains geographic coordinates for the facilities represented in the records, allowing their locations to be visualized. Figure 1 shows the distribution of facilities by facility type across the geographic area represented in the dataset.

The records also span from July 2024 through September 2026. Figure 2 shows the number of inspection-level observations recorded in each month during this period. Inspection counts vary from month to month, and the beginning and end of the study period represent partial calendar years.

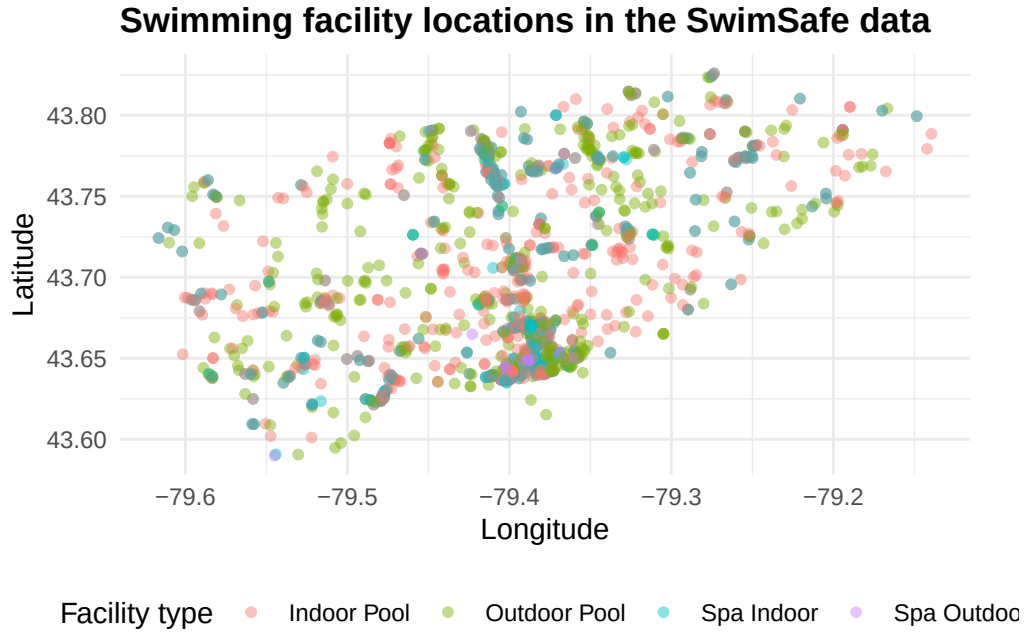


Figure 1: Geographic distribution of swimming facilities represented in the SwimSafe data.

The figure provides geographic context for the facilities included in the analysis; it is descriptive and is not used to evaluate geographic differences in inspection outcomes.

The inspection records also vary in their temporal coverage. Figure 2 shows the monthly number of inspection-level observations available during the study period.

Inspection counts vary from month to month. The dataset covers part of 2024, all of 2025, and part of 2026, so the beginning and end of the study period should not be interpreted as complete-year comparisons.

With the geographic and temporal coverage of the dataset established, the analysis next examines how inspection outcomes and recorded infraction patterns differ across facility types.

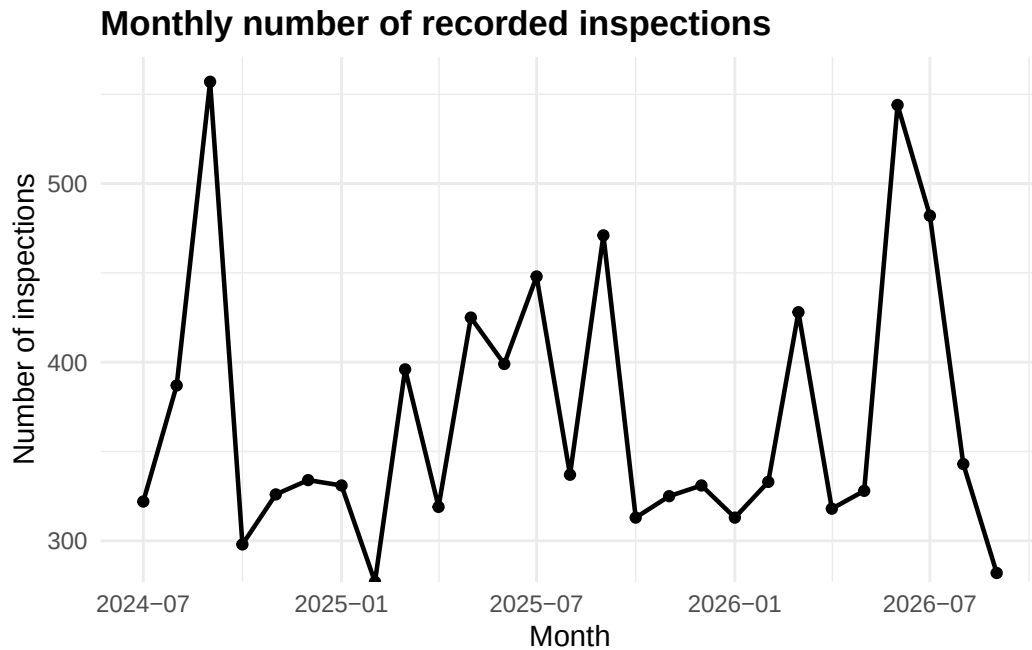


Figure 2: Monthly number of recorded inspections in the SwimSafe inspection-level dataset.

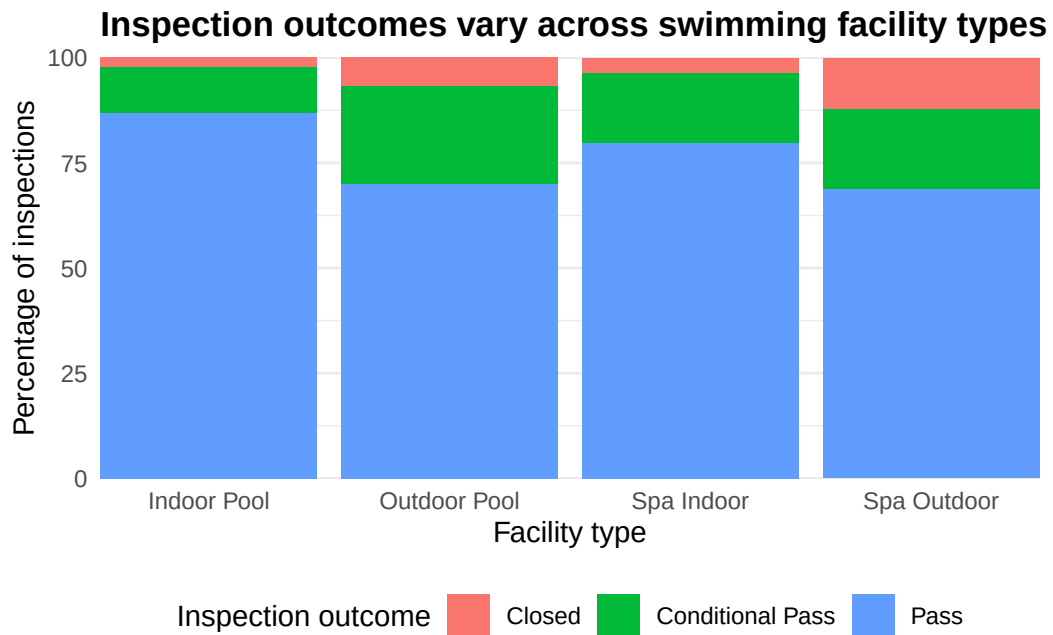


Figure 3: Inspection outcomes by swimming facility type.

3 Results

3.1 Inspection outcomes

Figure 3 shows the distribution of inspection outcomes across the four facility types. Indoor pools have the highest proportion of inspections recorded as Pass, at 86.9%, while Outdoor Pools and Spa Outdoor facilities have lower proportions of Pass inspections, at 69.9% and 68.7%, respectively. The proportion of inspections that were not recorded as Pass was therefore higher for outdoor facility types than for Indoor Pools. Spa Indoor facilities fall between these groups, with 79.7% of inspections recorded as Pass.

3.2 Recorded infraction types

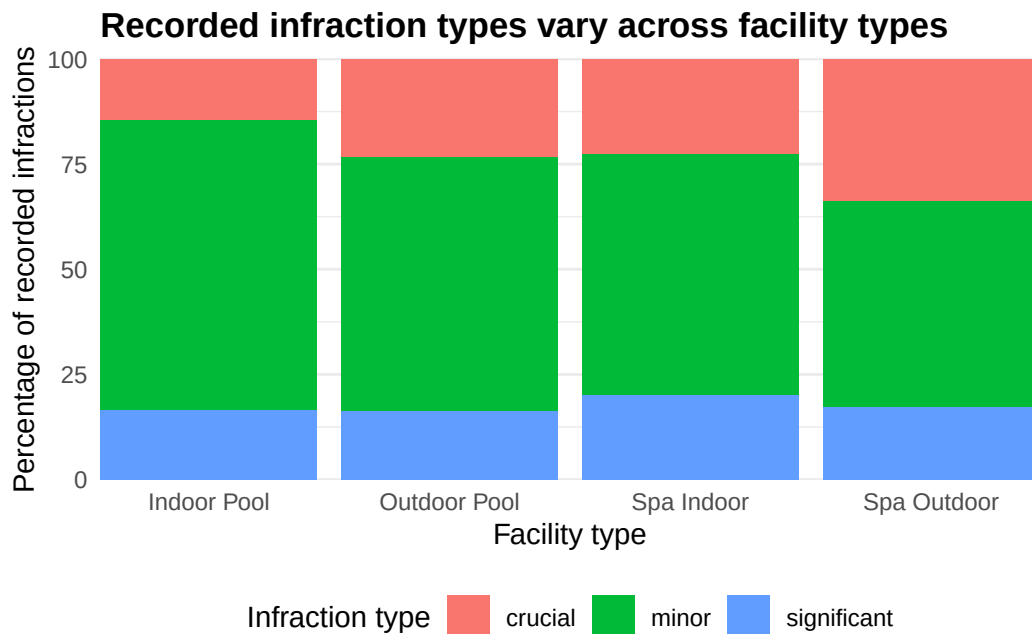


Figure 4: Distribution of recorded infraction types by swimming facility type.

Figure 4 shows that the composition of recorded infraction types differs across facility types. Indoor Pools have the largest share of recorded infractions classified as Minor, at 68.9%, whereas Spa Outdoor facilities have the smallest share, at 49.1%. The combined share of Significant and Crucial infractions ranges from 31.1% for Indoor Pools to 50.9% for Spa Outdoor facilities. These percentages describe the distribution of recorded infractions within each facility type; they do not represent the percentage of inspections with an infraction of a particular type.

3.3 Infraction categories

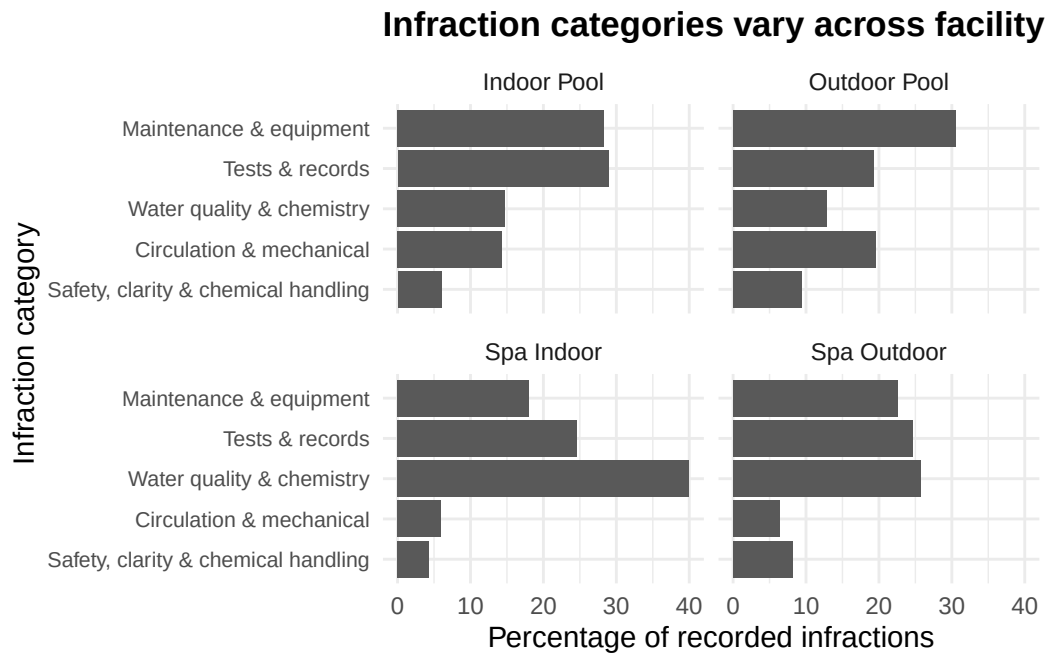


Figure 5: Main recorded infraction categories by swimming facility type.

Figure 5 shows differences in the categories of recorded infractions across facility types. For Indoor Pools, Tests & Records and Maintenance & Equipment account for the largest shares of recorded infractions. Outdoor Pools have a relatively large share of Maintenance & Equipment and Circulation & Mechanical infractions. In contrast, Water Quality & Chemistry is the largest category for Spa Indoor facilities. These patterns show that the distribution of recorded infraction categories differs across facility types.

3.4 Common specific deficiencies

Table 3: Three most frequently recorded specific deficiencies by facility type

Facility type	Specific deficiency	Number recorded
Indoor Pool	Pool/deck equipment and first aid kit maintenance	530
Indoor Pool	Pool alkalinity	355
Indoor Pool	Pool pH	303
Outdoor Pool	Pool/deck equipment and first aid kit maintenance	185
Outdoor Pool	Pool components not maintained in working order	161
Outdoor Pool	Pool pH	143
Spa Indoor	Spa alkalinity	477
Spa Indoor	Spa pH	375
Spa Indoor	Total bromine	266
Spa Outdoor	Spa pH	24
Spa Outdoor	Health hazard conditions	23
Spa Outdoor	Free available chlorine	20

The specific deficiencies provide further detail on the broader categories shown in Figure 5. Table 3 presents the three most frequently recorded specific deficiencies within each facility type. Among Indoor Pools, maintenance of pool and deck equipment and first aid kits is the most frequently recorded specific deficiency, followed by pool alkalinity and pool pH. For Spa Indoor facilities, spa alkalinity and spa pH are the most frequently recorded deficiencies. The most common deficiencies therefore differ across facility types, although some relate to water quality and maintenance across multiple types.

4 Discussion

4.1 Main findings

This analysis shows that inspection outcomes and recorded infraction patterns vary across Toronto's swimming facility types. Indoor Pools had the highest proportion of inspections recorded as Pass, while Outdoor Pools and Spa Outdoor facilities had higher proportions of non-Pass outcomes. The composition of recorded infractions also differed across facility types: Indoor Pools had larger shares of infractions related to Tests & Records and Maintenance & Equipment, Outdoor Pools had relatively larger shares related to Maintenance & Equipment

and Circulation & Mechanical Systems, and Spa Indoor facilities had a particularly large share related to Water Quality & Chemistry.

Taken together, these results suggest that swimming facility types have distinct inspection profiles. The differences are visible both in the overall inspection outcome and in the categories and types of infractions recorded during inspections. However, these descriptive patterns should not be interpreted as evidence that facility type itself causes differences in inspection outcomes or that one facility type is inherently safer or less safe than another.

4.2 Limitations

Several limitations should be considered when interpreting these results. First, the study period covers only part of 2024 and part of 2026, so the data should not be used to make complete-year comparisons across those years. Second, the analysis is descriptive and does not establish causal relationships between facility type and inspection outcomes or infractions. Third, the number of inspections differs substantially across facility types, with Spa Outdoor facilities having considerably fewer observations than Indoor Pools. Fourth, the analysis reflects conditions recorded during inspections and therefore does not necessarily represent conditions at a facility outside the inspection period. Finally, the categories Minor, Significant, and Crucial are used as recorded in the dataset; this analysis does not assign additional definitions or meanings to these categories beyond the available metadata.

4.3 Ethical Considerations

The analysis uses publicly available administrative data about swimming facilities rather than individual-level personal information. The analysis therefore focuses on aggregate patterns across facility types rather than identifying or evaluating individual people. Facility-level inspection records can nevertheless influence perceptions of particular establishments, so the results are presented descriptively and are not interpreted as evidence that any individual facility is unsafe or non-compliant outside the recorded inspection context.

5 Conclusion

This analysis examined how inspection outcomes and recorded infraction patterns vary across swimming facility types in Toronto. The results show differences in both the distribution of inspection outcomes and the composition of recorded infractions, with outdoor facility types having higher observed non-Pass rates than Indoor Pools and different distributions of infraction categories across facility types. The findings provide a descriptive view of the patterns present in Toronto's SwimSafe inspection records and illustrate how open administrative data can be used to examine differences across types of public facilities.

Appendix

A. Data Processing and Variable Definitions

The SwimSafe dataset contains multiple records for individual inspections because an inspection may include multiple observations or recorded infractions. The analysis therefore aggregates the raw records to an inspection-level dataset. An inspection is defined as a unique combination of establishment (**estName**), facility type (**type**), and inspection date (**insDate**). This aggregation produces 9,967 inspection-level observations from the 17,891 raw records.

For each inspection, the recorded inspection status is retained using the following rule when multiple statuses occur within an inspection: **Closed** takes precedence over **Conditional Pass**, which takes precedence over **Pass**. The number of recorded infractions is calculated from the **infType** variable, with separate counts for Minor, Significant, and Crucial infractions. The number of distinct non-None infraction categories is also recorded.

A.1 Key Variables

Table 4: Key variables used in the SwimSafe analysis

Variable	Description
estName	Establishment associated with the inspection
type	Type of swimming facility
insDate	Date of the inspection
insStatus	Recorded inspection status
infType	Recorded type of infraction
infCategory	Category of recorded infraction
defDesc	Description of the specific deficiency or observation

B. Additional Analysis Details

The inspection outcome analysis calculates the percentage of inspections within each facility type that were recorded as Pass, Conditional Pass, or Closed. Percentages are therefore calculated separately within each facility type rather than across the full dataset.

For recorded infraction types, the analysis excludes observations with **infType** = "None" and calculates the proportion of recorded Minor, Significant, and Crucial infractions within each facility type. These percentages describe the composition of recorded infractions and do not represent the proportion of inspections receiving a particular infraction type.

For infraction categories, observations with `infCategory = "None"` are excluded. The analysis identifies the five categories with the largest total number of recorded infractions and compares their percentage distribution across facility types. Specific deficiencies are summarized by identifying the three most frequently recorded `defDesc` values within each facility type.

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